Reg. No. | | | | | | | | | | |

[12EC52/13EC52]

FIFTH SEMESTER B.E/B.TECH EXAMINATION DECEMBER–2015

FUNDAMENTALS OF MICROELECTRONICS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	Define overdrive voltage. Derive the expressions for drain current (I_D) in three different regions of operation of an n-MOSFET.	**09**
	b.	With neat sketches, discuss the gate resistance reduction technique in MOSFETs.	**05**
2.	**a.**	Define back gate effect. Explain any two second order effects in n-MOSFET with necessary graphs and expressions.	**08**
	b.	Obtain an expression for output voltage and the small-signal gain of a MOSFET connected in common-source configuration with resistive load under $\lambda = 0$.	**06**
3.	**a.**	Derive an expression for gain in source follower and write a small-signal equivalent circuit of voltage buffer.	**07**
	b.	State and prove the Miller's theorem. Draw the large-signal characteristics of folded cascode circuit.	**07**
4	**a.**	Define symmetric differential pair. Discuss resistive mismatch in symmetric differential pair and obtain an expression for common mode gain and terminal voltages.	**10**
	b.	Explain the operation of a basic differential pair with current source as a load.	**04**
5.	**a.**	Discuss in detail, the frequency compensation of two-stage op-amp using miller's compensation technique.	**09**
	b.	Describe the PSRR and Slew Rate in two-stage operational-amplifiers.	**05**
6.	**a.**	Design a parallel switched capacitor network and derive an expression for equivalent resistance.	**08**
	b.	Explain aperture uncertainty in sample-hold circuit. Find the maximum sampling error for a S/H circuit that is sampling a sinusoidal input signal that could be described as Vin=A sin2πft where A is 2V and f=100KHz. Assume that the aperture uncertainty is equal to 0.5ns.	**06**

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7. a. Explain the charge scaling DAC with necessary circuit and expressions. Mention the limitations of charge scaling DAC. **07**

b. Design a 4-bit DAC using R-2R architecture with $R=2K\Omega$, $R_f=4K\Omega$ and $V_{ref}=6V$. Assume that the resistances of the switches are negligible. Determine the value of total current (i_{ont}) and the corresponding output voltages (V_{out}) for the digital inputs 1000, 1011, 0110, 1010, 1110. **07**

8. a. Discuss the working of pipeline ADC with a neat block diagram and show the conversion for 6.6V and 5V (3-bit, $V_{ref}=7V$) using the same type. **08**

b. Differentiate between the following ADC types: **06**

i. Single slope ADC and Dual slope ADC

ii. Flash ADC and two-step flash ADC